

Econ 131
Spring 2026
Danny Yagan

Problem Set 1

DUE: 02-25-2026 5PM PT on Gradescope

Student Name:

Student ID:

- Write or type your answers clearly and in dark ink (physical or electronic ink) so that your responses are legible.
- Tag each of your answers on Gradescope so that it is clear what responses are to which questions.
- **Although you may work in groups**, each student must submit individual sets of solutions. You must note the names of other students that you worked with. Write their names here:

1. True/False Statements [30 points]

Determine whether each statement is true, false, or uncertain and explain why. Answers with no explanation will receive no points. [5 points each]

- (a) Difference-in-differences yields the causal effect of interest if the treatment group and the control group have identical outcome levels before the treatment begins.

- (b) In January 2025, the Congressional Budget Office released its updated budget outlook. Download it [here](#). Look at the top half of Table B-1. Compute the average interest rate on government debt in 2035 r_{2035} by dividing 2035 Net Interest by 2034 Debt Held By The Public. Compute the 2035 GDP growth rate g_{2035} by comparing 2034 GDP to 2035 GDP. True/False/Uncertain: If the interest rate on government debt r_t and the GDP growth rate g_t stay constant forever at their 2035 levels, the government can run a permanent primary deficit while maintaining a stable debt to GDP level.

- (c) Adult income differences across U.S. local areas and across U.S. racial and ethnic groups are explained by differences in how rich or poor their parents were.

- (d) Suppose that a government applies a 10% income tax and uses the proceeds to give everyone a \$10,000 cash transfer. This tax-and-transfer system is progressive.

- (e) If a government places a \$1 per unit tax on producers of cigarettes rather than on consumers, producers will bear a larger economic burden of the tax than they would if the tax were placed on consumers instead.

- (f) When a sales tax is made more visible and salient to consumers at the point of purchase, it breaks the standard tax incidence result that equilibrium is independent of whether the tax is nominally placed on consumers or producers.

2. Difference-in-Differences (DiD) and Parallel Trends [30 points]

A policy is implemented in the **treated** group starting in **2015**. The **control** group is not directly affected. Let $Y_{g,t}$ denote the average outcome for group $g \in \{\text{Treated}, \text{Control}\}$ in year t .

Throughout, use:

Pre period = 2010–2014, Post period = 2015–2020.

The DiD estimand is:

$$\hat{\tau}_{DiD} = \left(\bar{Y}_{T, \text{Post}} - \bar{Y}_{T, \text{Pre}} \right) - \left(\bar{Y}_{C, \text{Post}} - \bar{Y}_{C, \text{Pre}} \right).$$

Scenario 1 [18 points]

Table 1: Scenario 1 outcomes, 2010–2020

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Control ($Y_{C,t}$)	50	51	52	53	54	55	56	57	58	59	60
Treated ($Y_{T,t}$)	55	56	57	58	59	62	63	64	65	66	67

- (a) [4 points] *Compute the four group means: $\bar{Y}_{T, \text{Pre}}$, $\bar{Y}_{T, \text{Post}}$, $\bar{Y}_{C, \text{Pre}}$, $\bar{Y}_{C, \text{Post}}$. Show your work.*

(b) [4 points] *Compute the changes within each group: $\Delta_T = \bar{Y}_{T,Post} - \bar{Y}_{T,Pre}$ and $\Delta_C = \bar{Y}_{C,Post} - \bar{Y}_{C,Pre}$.*

(c) [5 points] *Compute $\hat{\tau}_{DiD} = \Delta_T - \Delta_C$. In 1-2 sentences, explain what this number means.*

- (d) [5 points] *On the same axes, plot $Y_{T,t}$ and $Y_{C,t}$ for $t = 2010, \dots, 2020$. Draw a vertical line at 2015. Based only on 2010–2014, does parallel trends look plausible?*

Scenario 2 [12 points]

Table 2: Scenario 2 outcomes, 2010–2020 (same DiD as Scenario 1, but violates parallel trends)

Year	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Control ($Y_{C,t}$)	50	51	52	53	54	55	56	57	58	59	60
Treated ($Y_{T,t}$)	53	55	57	59	61	66	66	65	64	63	63

- (e) [5 points] Repeat the same steps as Scenario 1. Compute $\bar{Y}_{T,\text{Pre}}$, $\bar{Y}_{T,\text{Post}}$, $\bar{Y}_{C,\text{Pre}}$, $\bar{Y}_{C,\text{Post}}$, then Δ_T , Δ_C , and finally $\hat{\tau}_{DiD}$.

(f) [4 points] On the same axes, plot $Y_{T,t}$ and $Y_{C,t}$ for $t = 2010, \dots, 2020$. Draw a vertical line at 2015. Based only on 2010–2014, does parallel trends look plausible in Scenario 2?

(g) [3 points] Even if the average DiD number matches Scenario 1, why is Scenario 2 more concerning? (1–2 sentences.)

3. Tariffs and Welfare [28 points]

Figure 1 displays the case of a small open economy facing a world price and imposing a tariff.

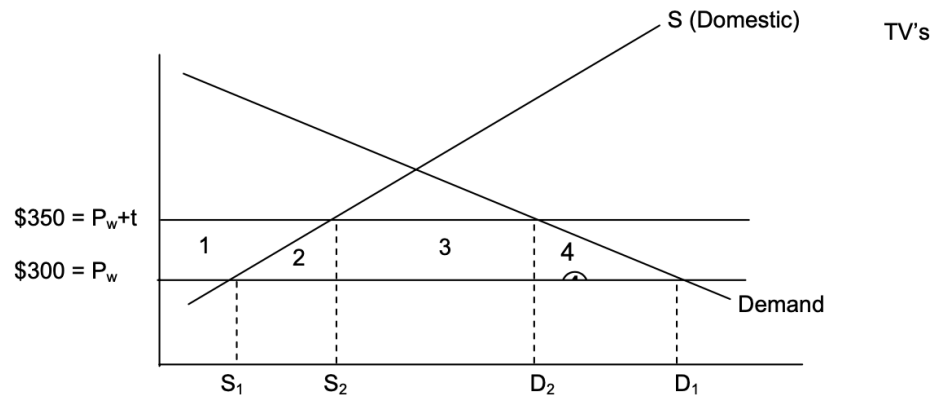


Figure 1: Small Country Case

- (a) [2 points] On the diagram, circle and label the world price pre-tariff, the tariff-inclusive domestic price, and draw a vertical bracket showing the size of the tariff wedge.

- (b) [3 points] Label the domestic quantity supplied and domestic quantity demanded both before and after the tariff. On the horizontal axis, mark the import quantity before the tariff and after the tariff.
- (c) [8 points] The lecture diagram labels four areas (1, 2, 3, 4). For each numbered area: shade/outline the area clearly, write what it represents (e.g. “producer surplus gain”), state who gets it (consumers, producers, government), and say whether it is a gain or loss for the home country. [2 points per area]

- (d) [3 points] Identify which areas are the deadweight loss triangles in the small-country case. Next to each triangle, write a short phrase explaining the economic distortion it captures: one triangle from too much domestic production (production distortion), and one triangle from too little domestic consumption (consumption distortion).

- (e) [8 points] Now consider the following specific example. A small country has domestic supply and demand curves for steel (quantities in thousands of tons):

$$Q^D = 100 - P$$

$$Q^S = P$$

where P is the price in dollars per ton. The world price of steel is $P_W = \$20$ per ton.

- i. [2 points] Without a tariff, compute the domestic quantity demanded, domestic quantity supplied, and imports. Show your work.

- ii. [2 points] Now suppose the government imposes a specific tariff of $t = \$10$ per ton on imported steel. Compute the new domestic price, domestic quantity demanded, domestic quantity supplied, and imports. Show your work.

- iii. [2 points] Calculate the following welfare effects (show your calculations): consumer surplus loss, producer surplus gain, government tariff revenue, and total deadweight loss.

- iv. [2 points] Verify that your deadweight loss equals the sum of the production distortion triangle and the consumption distortion triangle.

- (f) [4 points] Now consider a large country that *can* affect world prices. In 3–4 sentences, explain how the analysis would differ. What additional welfare effect arises when a large country imposes a tariff, and why does this make the overall welfare impact ambiguous?